



Section C: APIs, Tools & Developer Concepts

Research Tip: Search for 'What is an API for beginners' on YouTube or read documentation from platforms like Postman, RapidAPI, or Anthropic.

19. What is an API (Application Programming Interface)? Explain what it is using a simple real-world analogy. Why are APIs important in modern software and data systems? An API is a set of rules and protocols that allows different software applications to communicate and exchange data with each other. An API allows an application to request data or services without understanding the internal workings of the other system. API is important in modern software and data systems because it enables system integration & automation. We use many different systems. Suppose Real-Time Data Stream, API make it possible to exchange information instantly.

20. What is an API Key? What is its purpose? How should it be stored and protected, and what could happen if it is exposed publicly? An API Key is a unique secret code or token used to identify and authenticate an application when it accesses an API. Purpose of an API key server. Several important purposes: Authentication, authorization, usage tracking and billing and quotas. API keys should be treated like passwords. Best practices for storing API keys: secure vaults, environment variables, secret management. Exposed API keys can create serious security and financial risks.

21. What is a REST API? What does REST stand for and what are the main HTTP methods (GET, POST, PUT, DELETE) used in REST APIs? Briefly describe each method. A REST API is a type of API that allows applications to communicate over the Internet using standard HTTP methods. REST stands for Representational State Transfer. Main HTTP methods used in REST APIs: 1. GET 2. POST 3. PUT 4. DELETE

22. What is JSON (JavaScript Object Notation)? Why is it commonly used when working with APIs and data exchange? Write a simple example of a JSON object. JSON is a lightweight format for storing and exchanging data. It represents data using key-value pairs, making it easy for both humans and computers to read and understand. When an application sends or receives data through an API, the data needs to be in a format that both systems can understand, JSON is popular because it is

① Easy to Read ② Lightweight ③ Easy for computers to process. ④ Ideal for web and mobile applications



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The image is obtained from:

- Camera
- Smartphone
- A Scanner
- CCTV footage

Step 2

The image is broken down into thousand or millions of pixels. Each pixel contains information such as:

- Color
- Brightness
- Position

Step 3

AI algorithms look for important features such as

- Edges
- Shapes
- Corners
- Colors
- Textures

Step 4: Pattern Recognition

Using machine learning and deep learning, the system compares detected features with patterns learned during the training.

Step 5: Classification or Decision

The system produces a result. Real-world use case 1

Facial Recognition, A camera captures a person's face. The

system detects the face, measures facial features, compares

them with stored records and identifies the individual. Real

world use case 2: Self-driving cars. Camera continuously captures

images of the road, computer vision identifies: Traffic lights, Road

signs, Pedestrians, other vehicles, the vehicles then decide how

to respond



23. What is Prompt Engineering? Why is it an important skill when working with AI tools like ChatGPT or Claude? Provide two examples of a poorly written prompt and an improved version of each.

Prompt Eng. means it's the skill of creating clear, specific and well-structured prompts that help AI tools generate better results. AI tools generate responses based on the instructions they receive. The quality of the output often depends on the quality of the prompt.

Poor Prompt: Write an email.

Better Prompt: Write a professional email to a parent regarding a child's progress in school over within 2 business days.

24. What is a map function? Explain what it does in programming and data processing. Provide a simple example in either Python or JavaScript.

The `map()` function processes each element in a collection using a given function and produces a new collection with the updated values. Why is `map()` in Data Processing needed? Transform data, clean data, convert data types, apply calculations. Python Example: Suppose you have a list of school fees payments and want to add 15% VAT

Payments = [1000, 2000, 3000]

2. `Vat = payments * list(map(lambda x: x * 1.15, payments))`

3. `Print(Vat, payments)`

25. What is Cloud Computing? Name the three main cloud service models (IaaS, PaaS, SaaS) and briefly explain each. Give one example of a product for each model.

Cloud computing is a technology that allows individuals and organizations to access computing resources on demand through the internet, paying only for what they use. PaaS (Platform as a Service) provides a platform for developers to build, test and deploy applications without worrying about underlying infrastructure. Examples: Microsoft Azure App Service, AWS Elastic Beanstalk. SaaS (Software as a Service) provides fully functional software over the internet. Examples: Google Docs, Salesforce, Zoom.

26. What is a Version Control System? What is Git and why is it widely used by data analysts and developers? Briefly explain what a 'commit' and a 'branch' are.

Version Control System (VCS) is a tool that helps track changes made to files, documents, or source code over time. It allows multiple people to work on the same project while keeping a history. Git is a distributed version control system that allows developers and analysts to track, manage, and collaborate on changes to files and code.

Commit is a save record of changes made to files at a specific point in time. Think of a commit as saving a checkpoint in a game.

A branch is a separate version of a project where changes can be made without affecting the main project. Think of a branch as a copy of the project used for experimentation.



27. What is a Jupyter Notebook? What programming language is it most commonly used with? Why is it popular among Data Analysts and Data Scientists?

A Jupyter Notebook is an interactive computing environment that combines code, output, charts, and documentation in a single file. Jupyter Notebooks are most commonly used with Python. A notebook can contain code, results, graphs, notes and explanations. Interactive Data Analysis: users can run code one section at a time instead of running an entire program. It allows for data visualization. Jupyter Notebooks can display charts / graphs directly in the notebook.

28. What is Python? Why is it considered one of the most popular programming languages for Data Analytics and AI? Name four Python libraries commonly used in data work and briefly describe each.

Python is a versatile programming language used for software development, data analysis, automation, web development, Artificial Intelligence (AI) and machine learning. Python is easy to learn and read. Python uses a simple, human-readable syntax. Four common Python libraries used in data work. 1. Pandas is used for data manipulation and analysis. 2. NumPy! Provides support for numerical and mathematical operations. It is designed to handle: large datasets, Arrays, mathematical calculations. 3. Matplotlib is a data visualization library used to create charts and graphs. 4. Scikit-learn: is a machine learning library. It provides tools for predictive modeling, Classification, Clustering.

Pandas: Tables

NumPy: Numbers

Matplotlib: Charts

Scikit-learn: Predictions



Section D: Data Ethics, Governance & Careers

Research Tip: Explore the World Economic Forum, GDPR documentation, and LinkedIn Learning for content on data ethics and career pathways in data.

29. What is Data Privacy? Why is it important? Research the GDPR (General Data Protection Regulation) and explain what it is and which countries it applies to. Data Privacy is the practice of protecting personal information and giving individual control over how their data is collected, used and shared. Why Data Privacy is important: 1. Control data privacy enables help prevent, identify theft, fraud, unauthorized access, build trust and help meet legal requirements. GDPR applies throughout the European Union (EU) and its territories and can be incorporated into the European Economic Area making it applicable across the EEA.

30. What is Data Ethics? Provide three examples of unethical use of data or AI, and explain what makes each example problematic. Data Ethics is the practice of using data and AI responsibly, fairly, transparently, and in a way that respects people's privacy, rights and well-being. 1. Biased AI: A hiring AI company trained on AI hiring systems using historical employee data that mostly contained male candidates has one dominant gender group. Why is it unethical? Causes unfair discrimination, qualified candidates may be rejected, perpetuates historical bias, collecting personal data without consent, facial recognition for mass surveillance.

31. What is Data Governance? Define it and explain the key components of a good data governance framework. Data Governance is the process of managing data availability, quality, security, and usability to ensure that data is reliable and used responsibly across an organization. Key components of good Data Governance: 1. Data Quality management: This ensures that data is accurate, complete, timely, and fit for use. 2. Data Security: Protects data from unauthorized access, theft, or loss. 3. Data Privacy: Ensures that data is collected and used responsibly. 4. Data Ownership: Clarifies responsibility for data. 5. Data Lifecycle Management: Ensures data is managed throughout its lifecycle.

32. What is Bias in AI? Explain what it means for an AI model to be biased. Provide a real-world example of AI bias and describe the potential consequences. AI Bias is a systematic error that causes an AI model to perform poorly, or worse, to treat certain individuals or groups unfairly. AI model learn from historical data. 1. That data contains biases, errors, or unbalanced representation. The AI may learn and repeat these patterns. 2. AI hiring system: A company develops an AI tool to help screen job applicants. The model is trained on historical hiring data from past few years, historical: most successful candidates hired were men. Very few women were hired into certain roles.



22. What are the key differences between a Data Analyst, a Data Scientist, a Data Engineer, and an AI Engineer? Create a simple table comparing their roles, skills, and tasks.

Role	Primary Focus	Key Responsibilities	Key Skills
Data Analyst	Understanding historical data	Analyzing data, creating reports, data visualization	SQL, Excel, Power BI, data visualization, statistics
Data Scientist	Predictive analysis and insights	Build predictive models, machine learning	Statistics, machine learning, Python, data modeling
Data Engineer	Data infrastructure	Set up data pipelines, manage data storage, ensure data quality	Deep learning, NLP, machine learning, AI/ML
AI Engineer	AI Application Development	Develop AI models, deploy AI systems	

23. What is Business Intelligence (BI)? Name two popular BI tools (e.g., Power BI, Tableau) and explain how they help businesses make data-driven decisions. **Business Intelligence (BI)** refers to the technologies, tools, and practices used to transform raw data into meaningful insights that support decision-making. **Power BI** is a leading BI tool for visualizing data and reports. **Tableau** is a leading business intelligence and visualization tool that helps users explore data and create interactive reports.

Section E: Personal Reflection

Answer the following reflection questions thoughtfully and honestly. There is no right or wrong answer — we want to know what you think!

24. Which concept from this assignment did you find most interesting or surprising? Explain why.
The concept I found most interesting was Artificial Intelligence. The AI systems which are viewed as objective and unbiased, can actually make unfair decisions. They are trained on biased or incomplete data.

25. Which concept did you find most difficult to understand? What resources did you use to help you understand it better?



27. How do you see Data and AI affecting your future career or everyday life? Be specific with examples. Data and AI will have a significant impact on both my future career and my everyday life. As an AI Controller, I already work with large amounts of financial and customer data, and I believe AI will help me with my tasks faster, more accurately and more efficiently.

References

List all the sources you used to complete this assignment. Use the format below.

Author/Organization. (Year). Title of article or webpage. Website Name. URL

1. Data European Commission. (2022). Data (potential explained). European Commission. https://commission.europa.eu/innovation-topics/data-potential/data-potential_en.
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4. Python as Data Analyst. Third Edition. August 2022.
5. BrightLearn. Additional resources (A beginner's guide to cloud platforms for data analytics).
6. An open-source ESQL (introduction to SQL) by Iliya S. P. Oboznenko. 2014. 104 pages.

Good luck! We believe in you.

BrightLearn Data Analytics